

CSA0503-DATABASE MANAGEMENT SYSTEMS

EXPERIMENT 17 - HIGH LEVEL PROGRAMMING EXTENSIONS-FUNCTIONS

1)

```
mysql> CREATE FUNCTION GetCustomerLevel(  
->    p_creditlimit DECIMAL(10,2)  
-> )  
-> RETURNS VARCHAR(20)  
-> DETERMINISTIC  
-> BEGIN  
->     DECLARE customer_level VARCHAR(20);  
->  
->     IF p_creditlimit > 50000 THEN  
->         SET customer_level = 'PLATINUM';  
->  
->     ELSEIF p_creditlimit >= 10000 AND p_creditlimit <= 50000 THEN  
->         SET customer_level = 'GOLD';  
->  
->     ELSE  
->         SET customer_level = 'SILVER';  
->     END IF;  
->  
->     RETURN customer_level;  
-> END //
```

Query OK, 0 rows affected (0.01 sec)

```
mysql>  
mysql> DELIMITER ;  
mysql> SELECT GetCustomerLevel(75000) AS Customer_Level;
```

```
+-----+  
| Customer_Level |  
+-----+  
| PLATINUM      |  
+-----+  
1 row in set (0.01 sec)
```

```
mysql> SELECT GetCustomerLevel(25000) AS Customer_Level;
```

```
+-----+  
| Customer_Level |  
+-----+  
| GOLD          |  
+-----+  
1 row in set (0.00 sec)
```

```
mysql> SELECT GetCustomerLevel(5000) AS Customer_Level;
```

```
+-----+  
| Customer_Level |  
+-----+  
| SILVER        |  
+-----+  
1 row in set (0.00 sec)
```

2)

```
mysql> DELIMITER //
mysql>
mysql> CREATE PROCEDURE Factorial(
  ->     IN n INT,
  ->     OUT result BIGINT
  -> )
  -> BEGIN
  ->     DECLARE temp BIGINT;
  ->
  ->     IF n <= 1 THEN
  ->         SET result = 1;
  ->     ELSE
  ->         CALL Factorial(n - 1, temp);
  ->         SET result = n * temp;
  ->     END IF;
  -> END //
ERROR 1304 (42000): PROCEDURE Factorial already exists
mysql>
mysql> DELIMITER ;
mysql>
mysql> CALL Factorial(5, @result);
Query OK, 0 rows affected (0.00 sec)

mysql>
mysql> SELECT @result AS Factorial;
+-----+
| Factorial |
+-----+
|         120 |
+-----+
1 row in set (0.00 sec)

mysql> CALL Factorial(6, @result);
Query OK, 0 rows affected (0.00 sec)

mysql>
mysql> SELECT @result AS Factorial;
+-----+
| Factorial |
+-----+
|         720 |
+-----+
1 row in set (0.00 sec)
```

RESULT:

Thus the Functions using program in MySQL executed successfully .